Summary Notes: Database Security

• Database Security - Summary Notes**

• Target level: Primary (upper-primary)*

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Overview

• A **database** is a big digital filing cabinet that stores pictures, scores, stories and other information.

• **Security** is the set of rules and tools that keep the cabinet safe so that only the right people can see or change the data.

• Good security protects three things (the **CIA** triad, written in kid-friendly words):

| CIA | What it means for a database |

|-----|------------------------------|

| **Secret** (Confidentiality) | Only the right people can read the data. |

| **Correct** (Integrity) | The information stays accurate and unchanged unless a trusted person updates it. |

| **Ready** (Availability) | The data is there when we need it. |

• Think of a treasure chest that stays hidden, stays whole, and can be opened whenever the owner wants.*

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What a Database Is

• Stores many pieces of information, like a school notebook with separate sections.

• Can hold pictures, names, game scores, short stories, etc.

• Every piece has a label (a **name**) so the computer can find it quickly.

• Lives on a special computer called a **server** that is kept in a locked room.

• People can add, look at, or change data only if they have **permission**.

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Why We Need to Keep It Safe

• **Privacy:** If a stranger reads the data, private details (e.g., student names) can be exposed.

• **Accuracy:** Wrong changes make grades or scores incorrect.

• **Access:** If the database stops working, teachers and students can't get the information they need.

• Keeping the database safe stops cheating, stealing, or losing important work.

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Ways Bad Things Can Happen

• A teacher forgets to lock a password, letting others see the data.

• Someone sends a fake email (phishing) to trick a teacher into giving a password.

• Mistakes in the program create **weak spots** that let bad people in.

• A computer virus hides in a game and then harms the database.

• Too many kids try to open the same locker at once, causing the database to stop answering (a **Denial-of-Service** problem).

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Simple Ways to Protect the Database

• **Strong passwords** - secret codes that are hard to guess.

• **Limited keys (access control)** - only a few trusted people can change important data.

• **Regular updates (patches)** - fixing the lock as soon as a problem is found.

• **Secret coding (encryption)** - data is turned into a secret language while stored or sent, so even if someone grabs it, they cannot read it.

• **Locked server room** - the room where the server lives is kept locked and cool, like a school lab.

• **Watch-dog tools (monitoring)** - special programs watch for strange activity and sound an alarm if something looks wrong.

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Keeping a Spare Copy - Backups

• **What a backup is:** an extra copy of the whole database saved in a safe place, like a spare notebook kept at home.

• **Why backups matter:**

• If the main database breaks, is deleted, or is damaged by a virus, the backup lets us restore everything quickly.

• Storing the backup in a different location (e.g., another computer room or cloud storage) protects us from fire, flood, or theft that could destroy the original.

• **How backups are used:**

• Backups are made **regularly** (daily or weekly) so the newest information is always saved.

• Schools **test** their backups often to be sure the copy works when it is needed.

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Key Takeaways

• A database = digital filing cabinet for school and game information.

• Security protects **Secret**, **Correct**, and **Ready** data.

• Threats can come from careless insiders, tricksters, program bugs, viruses, or too many users at once.

• Simple defenses: strong passwords, limited keys, regular updates, encryption, locked rooms, and monitoring tools.

• **Backups** are spare copies that let us recover data fast if the original is lost or damaged.

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Conclusion

Understanding database security helps kids see why schools and games keep their information safe. By thinking of locks, secret codes, and spare copies, children can grasp how digital treasures stay private, correct, and available-building good habits for responsible digital citizenship.